

Rassel A Sadat

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SUMMARY

Curious and detail-oriented Computer Science graduate student with a strong interest in Data Intelligence, Artificial Intelligence, Machine Learning, and data-driven problem solving. Experienced in Python, machine learning, large language model (LLM) API integration, RESTful API development, data preprocessing, backend development, and MongoDB, building practical, reliable, full-stack solutions that transform raw information into structured data for intelligent systems and better decision-making. Quick to learn new technologies, skilled at solving complex problems, and focused on producing accurate, well-organized work. Eager to contribute analytical thinking, attention to detail, and a continuous learning mindset to AI-driven products and data-intensive workflows.

EDUCATION

VIT AP University, Amaravati, Andhra Pradesh <i>Integrated M.Tech in Computer Science Engineering</i> Current CGPA: 8.73	2022 – Present
Chinmaya Vidyalaya, Thrissur, Kerala Class XII (CBSE) Class X (CBSE)	2020 – 2022

EXPERIENCE

Flipkart GRID 6.0 – Software Development Track <i>Participant</i>	Aug 2024 – Sep 2024
- Collaborated in a national-level software engineering competition to solve real-world technical challenges under strict timelines. - Applied data structures, algorithms, and analytical thinking to design scalable and efficient software solutions. - Worked within structured development workflows while emphasizing clean implementation, performance optimization, and effective problem solving.	

PROJECTS

DocuCheck – AI-Powered Document Extraction & Verification System <i>Technologies: Python, FastAPI, React (Vite), SQLite, Google Gemini API, RESTful APIs</i>	2026
- Developed a full-stack AI document extraction system using FastAPI and React (Vite) that processes receipts and invoices through the Google Gemini multimodal API. - Designed an independent guardrail validation layer using deterministic arithmetic checks (line-item totals, subtotal and tax reconciliation) to verify AI-extracted data and flag inconsistencies. - Implemented a confidence-based review workflow flagging low-confidence extracted fields for human verification, and built an offline evaluation pipeline that achieved 93.3% field-level extraction accuracy against ground-truth data. - Structured persistent storage with SQLite to separately track raw AI output and human-corrected data for auditability.	
News Aggregator Platform <i>Technologies: MongoDB, Express.js, React, Node.js</i>	2024
- Developed a full-stack news aggregation platform that collected information from multiple sources using RESTful APIs. - Built backend services to validate, organize, and categorize incoming data into structured formats for efficient retrieval. - Implemented data filtering and preprocessing techniques to improve the quality and consistency of aggregated information. - Optimized backend processing and API workflows to improve response efficiency and data accessibility.	
AI-Powered Digital Twin Framework for Personalized Drug Response <i>Technologies: Python, Random Forest, Data Processing</i>	2025
- Developed an AI-powered digital twin framework using Python and Random Forest models to simulate patient-specific drug responses. - Cleaned, preprocessed, and prepared clinical datasets for machine learning workflows and predictive analysis. - Achieved a ROC-AUC score of 0.96 through feature engineering, model evaluation, and structured data preparation. - Generated visualizations and analytical insights to support data-driven decision making and model interpretation.	

SKILLS

Programming Languages: Python, Java, JavaScript, HTML, CSS

Data & AI: Data Collection, Data Cleaning, Data Preprocessing, Data Analysis, Structured Data, Semi-Structured Data, Machine Learning, Random Forest, Model Evaluation (ROC-AUC), Large Language Models (LLM), Google Gemini API, AI Integration, Data Validation

Backend & Web Development: Node.js, Express.js, FastAPI, RESTful APIs, API Integration, React (Vite), Modular Architecture

Databases: MongoDB, SQLite, NoSQL Data Handling

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Software Development Lifecycle (SDLC)

Tools: Git, GitHub, VS Code, Figma

LANGUAGES

Malayalam – Native | English – Professional | Hindi – Intermediate